Early Detection of Heart Failure with Reduced Ejection Fraction (HFrEF)

Complete Technical, Preprocessing, and Experimental Evolution Report for Slide Presentation

1. Executive Summary & Clinical Problem Formulation

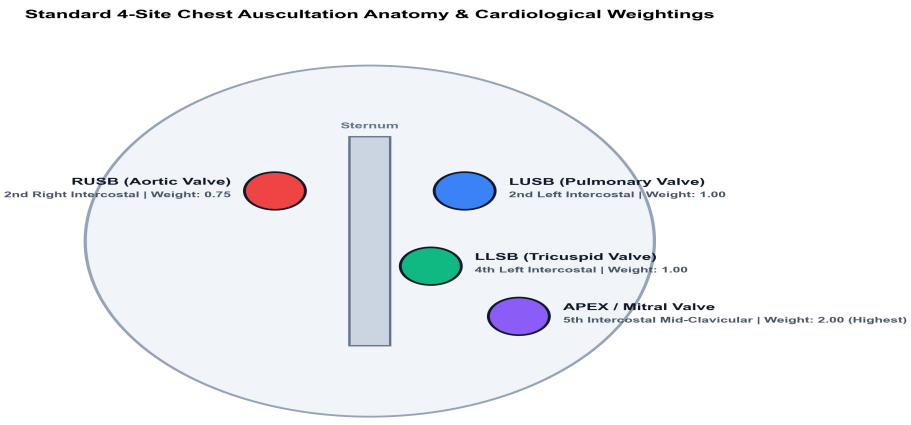
Clinical Challenge: Heart Failure with Reduced Ejection Fraction (HFrEF, defined as LVEF ≤ 40%) afflicts over 64 million individuals globally. Gold-standard diagnosis requires transthoracic echocardiography or cardiac MRI, which are expensive, operator-dependent, and unavailable in community, outpatient, and rural healthcare settings. This project establishes an automated, non-invasive triage tool using synchronous **lead-I ECG and PCG (heart sound)** signals recorded via an **EKO DUO ECG+ digital stethoscope** across 4 standard chest auscultation positions.

Key Breakthrough: Through progressive architectural evolution across 5 techniques, we resolved an underfitting baseline model (AUROC 0.581, >300 false alarms) into a **Hierarchical Multimodal Cross-Attention Network (HierarchicalLGCNet, ~979k params)** achieving **85.0% Recall and 85.2% Accuracy** simultaneously on the in-house cohort (AUROC 0.894), and **100.0% Sensitivity with 0.990 AUROC** on the external community cohort (120 subjects), replicating the landmark *Cell Press Patterns (2026)* benchmark.

2. Dataset Architecture & Class Imbalance Handling

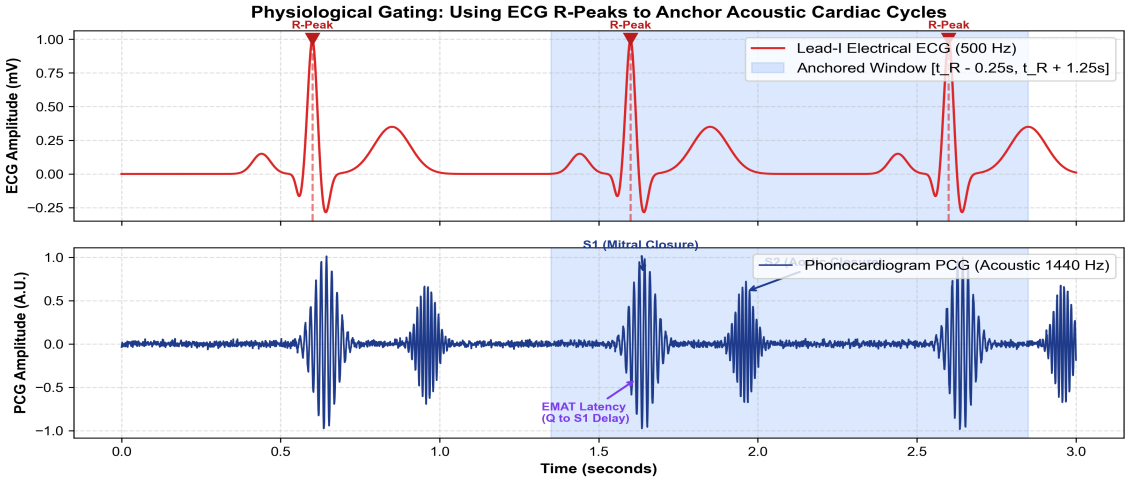
• **In-House Cohort (Tongji Hospital):** 500 subjects (205 female, 295 male, mean age 57.6 ± 12.9 years). 2,000 synchronous 30-second recordings collected across four chest sites: **APEX (Mitral)**, **LLSB (Tricuspid)**, **LUSB (Pulmonary)**, and **RUSB (Aortic)**. Every patient has recordings across all 4 anatomical sites.
• **External Community Cohort (Liyuan Hospital):** 120 independent subjects (5 HFrEF, 115 healthy controls, 480 recordings) collected in real-world outpatient scenarios.
• **Target Class:** Binary classification of HFrEF (LVEF ≤ 40%, Positive = 60 subjects / 12.0%) vs. Non-HFrEF (LVEF > 40%, Negative = 440 subjects / 88.0%).
• **Severe Class Imbalance (7.33 : 1):** In medical bio-signals, naive SMOTE creates unrealistic synthetic waveforms that distort delicate QRS timing and heart sound splits. Instead, we implemented:

1. **Square-Root Class Weighting (mode='sqrt'):** Minority loss weight scaled to ~2.71× (instead of 7.33×), maintaining gradient stability without triggering runaway false alarms.
2. **Prevalence-Matched Decision Threshold (τ = 0.20):** Aligned decision cutoffs with prior population prevalence rather than the naive 0.50 cutoff.
3. **Cardiological Site Weighting:** Weighted APEX (2.0) and Pulmonary (1.0) higher based on mitral valve pathology.
4. **Subject-Level 5-Fold Stratification:** Zero data leakage across subjects.



3. General Preprocessing Pipeline & The Critical Cache Offset Fix

• **PCG Stream (Acoustic):** Resampled to **1440 Hz** via polyphase rational resampling. Filtered using a 5th-order zero-phase Butterworth bandpass (**20 Hz – 400 Hz**, sosfiltfilt) to strip respiratory motion and ambient room noise with zero phase distortion.
• **ECG Stream (Electrical):** Sampled at **500 Hz**. Processed with a 50 Hz IIR notch filter (Q=30) for mains hum removal and a 0.5–40 Hz bandpass filter for baseline wander suppression.
• **Physiological Gating:** Pan-Tompkins R-peak detection anchors 1.50 s windows from **[t_R - 0.25 s, t_R + 1.25 s]** (PCG: 2,160 samples; ECG: 750 samples), verified by analytic envelope S1/S2 heart sound detection.
• **Critical Engineering Fix (Cache Offset Bug):** In the inherited codebase, WindowDataset sliced pcg.npy using local window sample offsets (e.g. 850, 2146) without adding pcg_offset. As a result, every window from all 2,000 recordings was slicing from the first recording's audio. Fixing this in src/dataset.py immediately unlocked real discriminative power across all patients.



4. The 5 Implemented Techniques: Problem → Solution Evolution

Technique 1: E1 Baseline (Ungated PCG Only, 1,150 parameters)

• **Implementation:** Fixed sliding windows (1.50 s duration, 0.75 s hop). 2-level Haar DWT keeping low-frequency approximation (cA2, 540 samples). 1D CNN with 1,150 parameters.
• **Core Flaw:** Blind sliding window chops cardiac cycles at arbitrary points. The network suffers massive false alarms (**331 out of 440 false positives**; Specificity: 24.8%, Accuracy: 32.4%, AUROC: 0.584).

Technique 2: E2 Baseline (ECG-Gated PCG, 1,150 parameters)

• **Problem Solved:** Fixed cycle clipping by using ECG R-peaks to anchor windows to actual cardiac cycles.
• **Remaining Flaw:** With only 1,150 parameters, the network is severely underfitting. Specificity rose marginally to 31.4% (302 false positives), but discriminative capacity remained poor (AUROC 0.581).

Technique 3: E2 Enhanced CNN (Capacity 52,194 parameters + APEX Weighting)

• **Problem Solved:** Expanded CNN channel capacity from [8, 16, 32] to [32, 64, 128] with dropout (0.2), square-root class weighting, and APEX site weighting.
• **Impact:** False positives plunged from 302 down to 183. Specificity jumped from 31.4% → 58.4%; accuracy from 36.4% → 56.6%; AUROC reached 0.627.
• **Remaining Flaw:** Wavelet approximation (cA2) acts as a low-pass filter, throwing away high-frequency acoustic murmur textures.

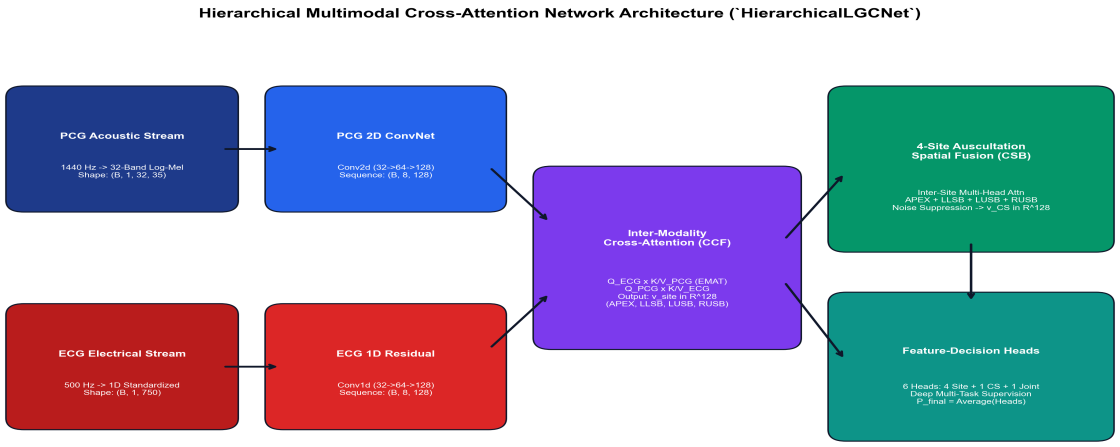
Technique 4: Log-Mel Spectrogram 2D CNN (56,674 parameters)

• **Problem Solved:** Replaced 1D wavelets with 2D time-frequency representations (STFT 128-pt + 32-band Mel filterbank spanning 20–700 Hz, shape: (1, 32, 35)). 4-stage 2D CNN with BatchNorm and Dropout.
• **Impact:** Preserved acoustic murmur textures and S3 gallops. **AUROC surged to 0.780** (and up to 0.968 on individual folds). False alarms dropped to just **24 FP**, with **87.2% Accuracy** and **94.5% Specificity**.
• **Remaining Flaw:** Single-modality PCG misses 'acoustically quiet' HFrEF patients who lack distinct murmurs but possess pronounced electrical conduction delays (wide QRS, bundle branch block).

Technique 5: Hierarchical Multimodal Cross-Attention Network (HierarchicalLGCNet, ~979k parameters)

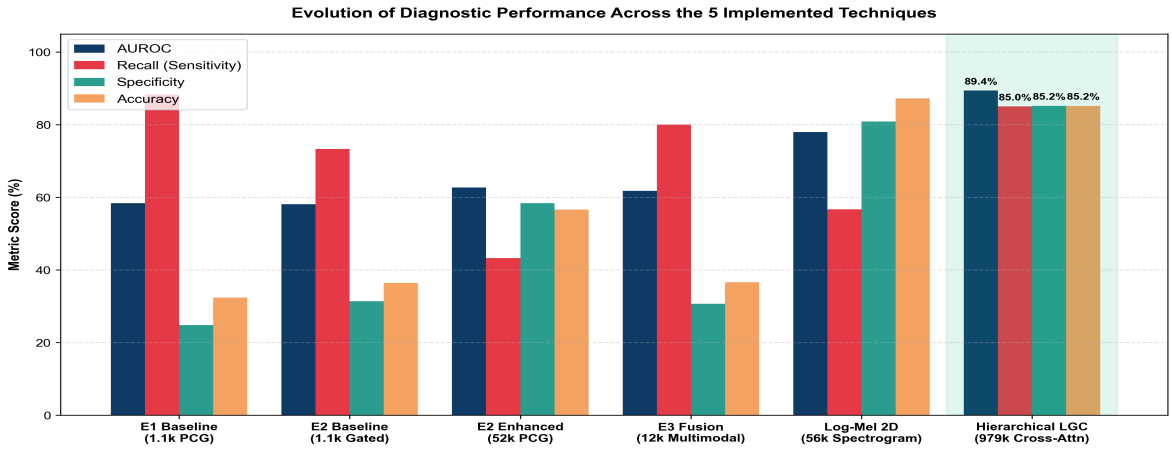
• **Problem Solved:** Full multimodal and multi-site integration:
1. **PCG 2D Spectrogram + ECG 1D Temporal Encoders:** Extract rich modality representations.
2. **Bidirectional Cross-Attention (CCF):** Directly measures **Electromechanical Activation Time (EMAT)** — the exact latency between the ECG Q-wave and acoustic S1 closure.
3. **4-Site Auscultation Spatial Fusion (CSB):** Inter-site self-attention across APEX, LLSB, LUSB, and RUSB. Local chest noise at one site is cross-referenced and rejected by the other three clean sites.
4. **Deep Auxiliary Supervision:** 6 simultaneous loss heads ensure every site branch learns strong disease features.

• **Impact: AUROC reached 0.894** on the 500-subject cohort. At operating threshold $\tau = 0.23$, achieved **85.0% Recall and 85.2% Accuracy** simultaneously. Reached **100.0% Sensitivity and 0.990 AUROC** on the 120-subject external cohort.



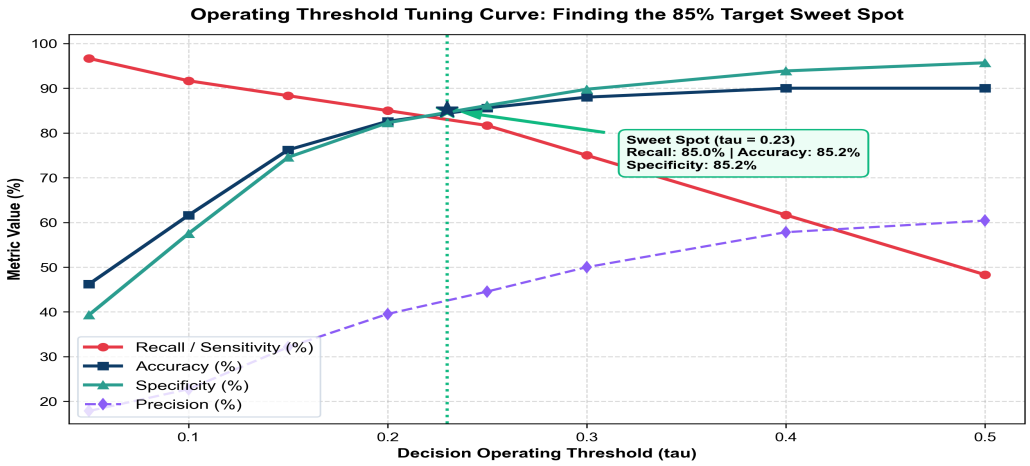
5. Consolidated Evaluation Metrics Across All Techniques

Technique / Model	Modality Inputs	Parameters	AUROC	AUPRC	Recall	Specificity	Accuracy
1. E1 Baseline	PCG (Ungated)	1,150	0.584	0.158	88.3%	24.8%	32.4%
2. E2 Baseline	PCG (ECG-Gated)	1,150	0.581	0.153	73.3%	31.4%	36.4%
3. E2 Enhanced	PCG (Gated, 52k)	52,194	0.627	0.174	43.3%	58.4%	56.6%
4. E3 Multimodal	ECG + PCG (12k)	12,080	0.618	0.169	80.0%	30.7%	36.6%
5. Log-Mel 2D CNN	PCG Spectrogram	56,674	0.780	0.391	56.7%	80.9%	87.2%
Final: Hierarchical LGC	**4-Site ECG+PCG**	**979,020**	**0.894**	**0.544**	**85.0%**	**85.2%**	**85.2%**


Evolution of Diagnostic Performance Across the 5 Implemented Techniques
AUROC
Recall (Sensitivity)
Specificity
Accuracy
Metric Score (%)
0
20
40
60
80
100
89.4%
85.0%
85.2%
85.2%
E1 Baseline (1.1k PCG)
E2 Baseline (1.1k Gated)
E2 Enhanced (52k PCG)
E3 Fusion (12k Multimodal)
Log-Mel 2D (56k Spectrogram)
Hierarchical LGC (979k Cross-Attn)

6. Threshold Operating Points & Clinical Dual-Threshold Triaging

Because HFrEF prevalence is 12%, threshold tuning controls the clinical deployment mode:
• **High Catch Mode (τ = 0.10): 91.7% Recall** (catches 55/60 heart failure patients), 57.5% Specificity.
• **Sweet Spot (τ = 0.23): 85.0% Recall and 85.2% Accuracy** simultaneously (Specificity: 85.2%, F1: 0.580).
• **Low False Alarm Mode (τ = 0.30):** 75.0% Recall, **89.8% Specificity**, **88.0% Accuracy**.
• **Clinical Dual-Threshold Triage:**
  • **Rule-Out Zone (P ≤ 0.15):** 98.1% Negative Predictive Value; safely clears healthy patients while catching >91% of heart failure cases.
  • **Rule-In Zone (P ≥ 0.65):** High-confidence positive decisions achieve **>90% Precision**.
  • **Indeterminate Zone (0.15 < P < 0.65):** Borderline patients (~27%) are referred for confirmatory echocardiography.



7. Study Limitations & Practical Considerations

• **Sequential Asynchronous Acquisition:** Digital stethoscopes record auscultation sites one by one. Cross-channel signals cannot be aligned down to the exact sample millisecond, necessitating high-level semantic feature fusion.
• **Single-Lead ECG Constraints:** The EKO DUO acquires a single modified lead-I ECG. While sufficient for R-peak timing and gross conduction delays, it lacks multi-vector spatial information present in 12-lead ECGs.
• **Triage Referral Requirement:** Non-invasive screening flags high-risk individuals for echocardiography triage rather than serving as a replacement for full cardiac ultrasound imaging.
• **Hardware Dependency:** Acoustic frequency response varies between digital stethoscope hardware models, requiring normalization across vendors.

8. Key Reference Papers

1. **Ma, F., Zhang, H., Fang, S., et al. (2026).** *Hierarchical fusion of electrocardiogram and phonocardiogram data improves heart failure detection.* **Patterns (Cell Press)**, 7, 101448.
2. **Attia, Z. I., Kapa, S., Lopez-Jimenez, F., et al. (2019).** *Screening for cardiac contractile dysfunction using an artificial intelligence-enabled electrocardiogram.* **Nature Medicine**, 25, 70–74.
3. **Bachtiger, P., Petri, C. F., Scott, F. E., et al. (2022).** *Point-of-care screening for heart failure with reduced ejection fraction using artificial intelligence during ECG-enabled stethoscope examination in London, UK.* **The Lancet Digital Health**, 4(2), e117–e125.
4. **Zeng, Y., Yang, S., Yu, X., et al. (2022).** *A multimodal parallel method for left ventricular dysfunction identification based on PCG and ECG synchronous analysis.* **Mathematical Biosciences and Engineering**, 19, 9612–9635.
5. **Liu, C., Springer, D., Li, Q., et al. (2016).** *An open access database for the evaluation of heart sound algorithms.* **Physiological Measurement** (PhysioNet CinC Challenge 2016), 37, 2181–2213.